

version command:

```
osfy@ubuntu-22-04-1-lts:~/Desktop$ java -version

openjdk version "11.0.17" 2022-10-18
OpenJDK Runtime Environment (build 11.0.17+8-post-Ubuntu-
1ubuntu222.04)
OpenJDK 64-Bit Server VM (build 11.0.17+8-post-Ubuntu-
1ubuntu222.04, mixed mode, sharing)
```

Verify the Maven version using the `mvn -version` command:

```
osfy@ubuntu-22-04-1-lts:~/Desktop$ mvn -version

Apache Maven 3.6.3
Maven home: /usr/share/maven
Java version: 11.0.17, vendor: Ubuntu, runtime: /usr/lib/jvm/
java-11-openjdk-amd64
Default locale: en_IN, platform encoding: UTF-8
OS name: "linux", version: "5.15.0-53-generic", arch:
"amd64", family: "unix"
```

We are going to use PetClinic as a sample application. PetClinic is a Spring Boot application built using Maven or Gradle. In this article we will build a JAR file and run it from the Docker container. You can visit <https://github.com/spring-projects/spring-petclinic> to know more.

Copy the GitHub repository URL from the above URL and use the `git clone` command to clone it locally.

```
osfy@ubuntu-22-04-1-lts:~/Desktop$ git clone https://github.
com/spring-projects/spring-petclinic.git
```

```
Cloning into 'spring-petclinic'...
remote: Enumerating objects: 9424, done.
remote: Total 9424 (delta 0), reused 0 (delta 0), pack-reused
9424
Receiving objects: 100% (9424/9424), 7.64 MiB | 1.55 MiB/s,
done.
Resolving deltas: 100% (3567/3567), done.
```

Once source code is available, go to the root of the project using the `change directory` command as given below, and execute the `./mvnw package` command to create a JAR file. It will try to download multiple dependencies and hence will take some time to complete execution.

```
osfy@ubuntu-22-04-1-lts:~/Desktop$ cd spring-petclinic
osfy@ubuntu-22-04-1-lts:~/Desktop/spring-petclinic$ ./mvnw
package
Warning: JAVA_HOME environment variable is not set.
[INFO] Scanning for projects...
```

```
Downloading from spring-snapshots: https://repo.spring.
io/snapshot/org/springframework/boot/spring-boot-starter-
parent/2.7.3/spring-boot-starter-parent-2.7.3.pom
Downloading from spring-milestones: https://repo.spring.
io/milestone/org/springframework/boot/spring-boot-starter-
parent/2.7.3/spring-boot-starter-parent-2.7.3.pom
Downloading from central: https://repo.maven.apache.
org/maven2/org/springframework/boot/spring-boot-starter-
parent/2.7.3/spring-boot-starter-parent-2.7.3.pom
```

```
.
.
.
Downloaded from central: https://repo.maven.apache.
org/maven2/org/springframework/session/spring-session-
bom/2021.2.0/spring-session-bom-2021.2.0.pom (3.1 kB at 6.0
kB/s)
[INFO] -----< org.springframework.samples:spring-
petclinic >-----
[INFO] Building petclinic 2.7.3
[INFO] -----[ jar ]-----
Downloading from spring-snapshots: https://repo.spring.io/
snapshot/org/springframe
```

```
Downloaded from central: https://repo.maven.apache.org/
maven2/org/codehaus/plexus/plexus-io/3.2.0/plexus-io-
3.2.0.jar (76 kB at 28 kB/s)
```

```
Downloaded from central: https://repo.maven.apache.org/
maven2/org/codehaus/plexus/plexus-archiver/4.2.7/plexus-
archiver-4.2.7.jar (195 kB at 69 kB/s)
```

```
Downloaded from central: https://repo.maven.apache.org/
maven2/org/apache/commons/commons-compress/1.20/commons-
compress-1.20.jar (632 kB at 218 kB/s)
```

```
Downloaded from central: https://repo.maven.apache.org/
maven2/org/tukaani/xz/1.9/xz-1.9.jar (116 kB at 36 kB/s)
```

```
Downloaded from central: https://repo.maven.apache.org/
maven2/org/codehaus/plexus/plexus-utils/3.3.1/plexus-utils-
3.3.1.jar (262 kB at 70 kB/s)
```

```
[INFO] Building jar: /home/osfy/Desktop/spring-petclinic/
target/spring-petclinic-2.7.3.jar
[INFO]
[INFO] --- spring-boot-maven-plugin:2.7.3:repackage
(repackage) @ spring-petclinic ---
```

Check the `.m2` folder available in the `Home` directory. It will have all the dependencies downloaded in the mvn package execution, as seen in Figure 3.